Lab program 3:

Code a Dockized python flask or node.js application

Project structure:

Dockerfile

app.py

requirements.txt

Step 1: create Dockerfile

sudo nano Dockerfile

Dockerfile

```
FROM python:latest

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python","app.py"]
```

Step 2: create requirements.txt

sudo nano requirements.txt

requirements.txt

```
flask==2.3.3
```

Step 3: create app.py

sudo nano app.py

app.py

from flask import flask

app=Flask(__**name**__)

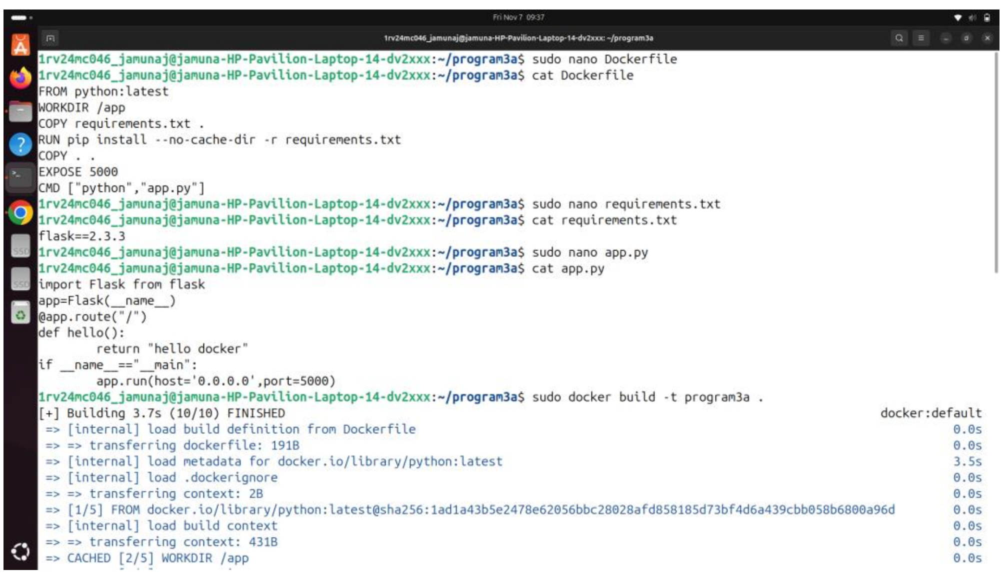
@app.route("/")

def hello():

return "hello docker"

if __**name**__=="__main__":

app.run(host='0.0.0.0',port=5000)



Step 4: Build the image

docker build –t program3 .

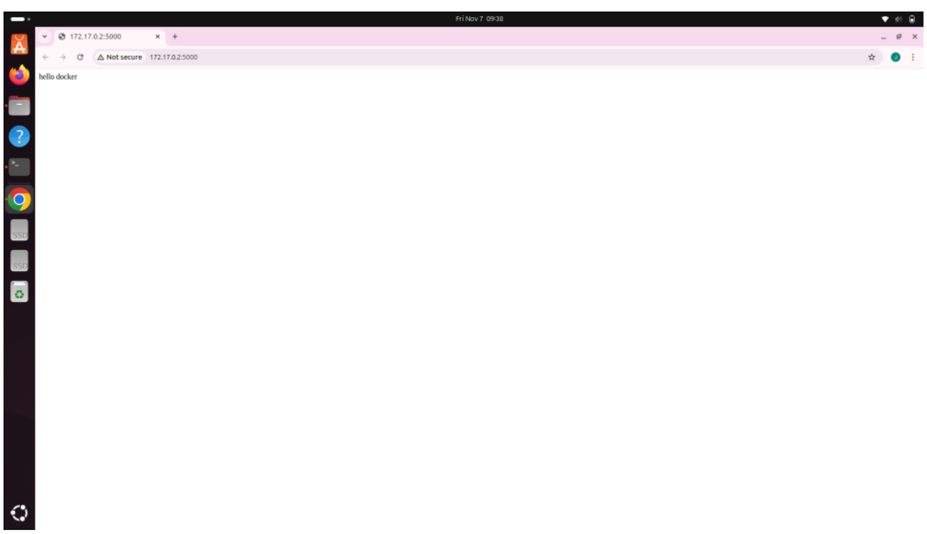
```
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/program3a$ sudo docker build -t program3a .
[+] Building 3.7s (10/10) FINISHED                                                              docker:default
 => [internal] load build definition from Dockerfile                                                      0.0s
 => => transferring dockerfile: 191B                                                                      0.0s
 => [internal] load metadata for docker.io/library/python:latest                                          3.5s
 => [internal] load .dockerignore                                                                         0.0s
 => => transferring context: 2B                                                                           0.0s
 => [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2478e62056bbc28028afd858185d73bf4d6a439cbb058b6800a96d  0.0s
 => [internal] load build context                                                                         0.0s
 => => transferring context: 431B                                                                         0.0s
 => CACHED [2/5] WORKDIR /app                                                                             0.0s
 => CACHED [3/5] COPY requirements.txt .                                                                  0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt                                       0.0s
 => [5/5] COPY . .                                                                                        0.0s
 => exporting to image                                                                                    0.0s
 => => exporting layers                                                                                   0.0s
 => => writing image sha256:998e695078d5c8caf791c60553540cd356a923e8759525de69e34ce851e487ea              0.0s
 => => naming to docker.io/library/program3a                                                              0.0s
```

Step 5: run and assign the port number

docker run –p 5000:5000 program3

```
1rv24mc046_jamunaj@jamuna-HP-Pavilion-Laptop-14-dv2xxx:~/program3a$ sudo docker run -p 5000:5000 program3a
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [07/Nov/2025 04:02:34] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [07/Nov/2025 04:02:34] "GET /favicon.ico HTTP/1.1" 404 -
```

Step 6: open the browser http://localhost:5000 to see the output



Step 7: to remove the container first stop that container

docker ps –a

docker container stop program3

docker container rm program3a